

Widget Data Contract

One question per widget: what will the frontend ask you for, and what shape must the answer be?

HOW THE 37 TYPES ASK TODAY

3 send an aggregated query 34 ask for records

Every row in this document is read out of the code that builds the query - the same `queryFor` the running application calls - so nothing here is an intention. Read the publication contract first for what a Dataset must *declare*; this is about what gets *asked* afterwards.

1 · The request

One shape, for every widget.

```
interface DatasetQuery {  
  /** Group by these Dimensions. Omitted or empty means a single aggregate row. */  
  dimensions?: string[]  
  measures?: { field: string; aggregation: Aggregation }[]  
  timeRange?: { field: string; from?: string; to?: string; granularity?: TimeGranularity }  
  /** Keyed by Field key. Only Fields the Dataset declared `filterable`. */  
  filters?: Record<string, string | number>  
  sort?: { field: string; direction: 'ascending' | 'descending' }[]  
  limit?: number  
}  
  
type Aggregation = 'sum' | 'average' | 'count' | 'minimum' | 'maximum' | 'distinct-count'  
type TimeGranularity = 'day' | 'week' | 'month' | 'quarter' | 'year'
```

Aggregation is asked for, never applied afterwards. A widget requests "sum of revenue by region" and expects grouped, aggregated rows. It must not receive raw records and reduce them, for two reasons: it would let each widget decide what "sum" means, and it would ship records a Viewer may not be entitled to see.

An **aggregation** we send is always one the Dataset declared for that Measure. If we ask for something undeclared, that is our bug — reject it.

2 · The response

Four outcomes, and they must be distinguishable. This is the easiest part of the contract to get wrong, because nothing in the requirements says it about the API - only about the display.

```

type RetrievalOutcome =
  | { kind: 'rows'; rows: DatasetRow[]; totalCount: number }
  | { kind: 'empty' }      // authorized, and the Dataset has nothing to say
  | { kind: 'denied' }     // not authorized for this Dataset
  | { kind: 'withdrawn' }  // the Dataset is gone

type DatasetRow = Record<string, string | number | null>

```

A response of `[]` cannot carry this: empty, denied and withdrawn all look identical, and the frontend would have to guess. It renders each of the four differently — a denial says access was denied rather than showing zero, because showing zero teaches a Viewer the figure *is* zero.

A genuine failure is an error, not an outcome — the absence of an answer rather than one of the answers.

`kind: 'rows'` with an empty `rows` array is treated as a **contract breach** and surfaced as a failure, not as `empty`. Send `{ kind: 'empty' }`.

Row shape

Flat objects, keyed by `Field` key. An aggregated Measure comes back under its own field name — ask for `sum(revenue)` and the key is `revenue`, not `revenue_sum`. Values are `string`, `number` or `null`; dates are ISO strings (2026-08-07, or 2026-08 at month grain).

3 · Aggregation grain - the thing to settle with us

Of the 37 built widget types, **3 currently send an aggregated query and 34 ask for records**. That is not a recommendation, it is a report - and it needs a conversation before it meets a real database.

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The reason is our fixtures: most are already stored at the grain the chart draws. `sales-by-region` is one row per region, so a bar chart over it asks for records and gets exactly the six bars it wants. Against a raw table the same widget would pull every transaction and group in the browser, which is both slow and the thing §2 of the publication contract forbids.

What this means for you:

- The 3 single-aggregate types are correct as they stand and safe against any volume.
- For the rest, tell us where your data's grain sits. If a Dataset is already aggregated to the grain we draw, records are fine and the `rows` count in §4 is what to expect. If it is raw, we need to send `dimensions + measures` and we will fix the query — the mapping already carries the information, so it is our change, not yours.

- The row counts in §4 are what our *fixtures* return. Treat them as the volume a widget can usefully draw, not as a limit you should enforce: a calendar heatmap wants 365 points and a stat card wants 1.

This is tracked on our side as divergences **D13** and **D14** in `DIVERGENCES.md`.

4 · Every widget, and what it asks for

Needs is what a Dataset must offer for the widget to be *offered* at all - the Field roles, and how many of each. A widget whose required slots cannot be filled is never shown to an Author, so a Dataset missing a Time Dimension simply does not produce trend charts.

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Single value single-value

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
stat-card Stat card	Measure — 1 × Measure	sum(revenue)	SINGLE AGGREGATE	1
sparkline-card Sparkline card	Measure — 1 × Measure Period — 1 × Time Dimension	order by date ↑	RECORDS	365
delta-card Delta card	Measure — 1 × Measure Period — 1 × Time Dimension	order by month ↑	RECORDS	24
progress-tracker Progress tracker	Actual — 1 × Measure Target — 1 × Measure	order by month ↑	RECORDS	24

Trend trend

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
line-chart Line chart	Time axis — 1 × Time Dimension Measures — 1–5 × Measure	order by month ↑	RECORDS	24
area-chart Area chart	Time axis — 1 × Time Dimension Measures — 1–5 × Measure	order by date ↑	RECORDS	365
spline-chart Spline chart	Time axis — 1 × Time Dimension Measures — 1–5 × Measure	order by month ↑	RECORDS	24
step-chart Step chart	Time axis — 1 × Time Dimension Measures — 1–5 × Measure	order by month ↑	RECORDS	24

Categorical comparison categorical-comparison

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
bar-chart-vertical Bar chart	Categories — 1 × Dimension or Time Dimension Measures — 1–4 × Measure	no aggregation — all records	RECORDS	6
bar-chart-horizontal Horizontal bar	Categories — 1 × Dimension or Time Dimension Measures — 1–4 × Measure	no aggregation — all records	RECORDS	12
grouped-bar-chart Grouped bar	Categories — 1 × Dimension or Time Dimension Measures — 2–4 × Measure	no aggregation — all records	RECORDS	6
stacked-bar-chart Stacked bar	Categories — 1 × Dimension or Time Dimension Measures — 2–4 × Measure	no aggregation — all records	RECORDS	6

Composition composition

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
pie-chart Pie chart	Segments — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	6
donut-chart Donut chart	Segments — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	6
treemap Treemap	Rectangles — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	12

Ranking & flow
ranking-and-flow

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
ranked-list Ranked list	Entries — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	20
leaderboard Leaderboard	Entries — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	12
funnel Funnel	Stages — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	6
sankey Sankey	From — 1 × Dimension To — 1 × Dimension Volume — 1 × Measure	no aggregation — all records	RECORDS	10

Status
status

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
status-list Status list	Entities — 1 × Dimension State — 1 × Dimension Measure — up to 1 × Measure	no aggregation — all records	RECORDS	9
status-indicator Status tile	Entities — 1 × Dimension State — 1 × Dimension	no aggregation — all records	RECORDS	9
threshold-indicator Threshold indicator	Measure — 1 × Measure	average(uptime)	SINGLE AGGREGATE	1
alert-banner Alert banner	Measure — 1 × Measure	average(errorRate)	SINGLE AGGREGATE	1

Radial
radial

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
gauge Gauge	Actual — 1 × Measure Target — 1 × Measure	order by month ↑	RECORDS	24
radar-chart Radar chart	Entities — 1 × Dimension Measures — 3–8 × Measure	no aggregation — all records	RECORDS	12

Tabular tabular

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
data-table Data table	Columns — 1–12 × Dimension or Time Dimension or Measure	no aggregation — all records	RECORDS	120
pivot-table Pivot table	Rows — 1 × Dimension Columns — 1 × Dimension Measure — up to 1 × Measure	no aggregation — all records	RECORDS	120

Distribution distribution

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
histogram Histogram	Measure — 1 × Measure	no aggregation — all records	RECORDS	2000
box-plot Box plot	Groups — 1 × Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	2000

Correlation correlation

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
scatter-plot Scatter plot	Measures — 2 × Measure	no aggregation — all records	RECORDS	12
bubble-chart Bubble chart	Measures — 3 × Measure	no aggregation — all records	RECORDS	12

Temporal pattern temporal-pattern

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
calendar-heatmap Calendar heatmap	Date — 1 × Time Dimension Measure — 1 × Measure	order by date ↑	RECORDS	365
cohort-grid Cohort grid	Cohorts — 1 × Time Dimension Elapsed period — 1 × Dimension or Time Dimension Measure — 1 × Measure	no aggregation — all records	RECORDS	78
timeline-chart Gantt chart	Rows — 1 × Dimension Start and end — 2 × Measure Grouping — up to 1 × Dimension Progress — up to 1 × Measure	no aggregation — all records	RECORDS	9

Chronological chronological

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
activity-feed Activity feed	When — 1 × Time Dimension Actor — 1 × Dimension Action — 1 × Dimension	no aggregation — all records	RECORDS	60
event-log-view Event log	When — 1 × Time Dimension Columns — 1–8 × Dimension or Time Dimension or Measure	no aggregation — all records	RECORDS	60

Geospatial geospatial

TYPE	NEEDS	REQUEST TODAY	GRAIN	ROWS
point-map Point map	Place — 1 × Dimension, geographic Latitude — 1 × Measure, geographic Longitude — 1 × Measure, geographic Measure — 1 × Measure	no aggregation — all records	RECORDS	20

5 · Field roles, for reference

ROLE	WHAT IT IS	NOTES
dimension	Identifies or categorizes a record	

time-dimension	A Dimension whose values are points in time	Counts as a Dimension wherever one is required
measure	Can be meaningfully aggregated	Must declare which aggregations are meaningful

Two things a Dataset declares that change what we can offer:

- **filterable gates every filter.** A Viewer-facing filter or a Dashboard Control can only act on a Field you marked filterable. We enforce this twice — the picker will not offer it, and the query builder drops it — so an unmarked Field is unreachable by design.
- **semantic unlocks four Families.** Role says what a Field *is* and cannot say what it is *for*. A point map needs to know which Measure is a latitude; without it, a ranked list will happily rank countries by how far north they are. See `PUBLICATION_CONTRACT.md` for the values.

6 · What we will not ask you for

- **Writes.** Every widget is read-only, enforced by the props carrying no callback — there is nothing a widget *could* call to mutate.
- **Joins.** A widget draws from exactly one Dataset. If two Datasets need relating, that is a Dataset the Source System publishes, not a query we send.
- **Formatting.** Send numbers as numbers. `format` on a Field tells us how to render them.
- **Anything a Viewer may not see.** Authorization is resolved per Dataset per call, on your side. We ask; we do not carry a rule about who may read what.